

i3 SmartLabs & AI Platform — Enterprise Master Plan

Complete Technical Architecture, 45-Vendor Matrix, Multi-Cloud Federation & Commercial Model

Leadership: Senior Business Consultant & VP, i3 Technologies **Ecosystem:** IBM Cloud ROKS 4.17 · Red Hat AAP · Hybrid Edge **Planning Horizon:** Q4 2026 – Q4 2029

Ksh 154.95M YEAR 1 GROSS REVENUE	45 Vendors OPERATIONAL LAB MATRIX	USD 36K / Yr IBM CLOUD CREDITS (3 ACCOUNTS)	2,500 CONCURRENT STUDENT CAPACITY
--	---	---	---

1. EXECUTIVE STRATEGIC INTENT & COMMERCIAL THESIS

Instrumented, Interconnected & Intelligent Technologies Limited (i3 Technologies) is a premier East African systems integrator, enterprise solutions architect, and technical workforce training provider headquartered in Nairobi, Kenya. Our operational baseline includes an enterprise production deployment across 10 namespaces on IBM Cloud Red Hat OpenShift on IBM Cloud (ROKS 4.17 in Frankfurt), powering **EvalOS** (AI assessment & certification platform), **AfroERP** (ERPNext v15 with integrated KRA eTIMS & Safaricom Daraja M-Pesa), **Engage** (omnichannel SMS/WhatsApp CRM), **PMaaS** (Political Management as a Service), and **JupyterHub / Sage AI Lab** backed by an on-cluster LiteLLM gateway with local quantized LLMs (Granite 3.1, Qwen 2.5 7B/14B, Qwen Coder, and LLaVA Vision).

This blueprint formalizes the launch of **i3 SmartLabs**: a unified, multi-tenant Lab-as-a-Service (LaaS), Exam-as-a-Service (EaaS), and AI Sandbox commercial engine. By packaging 45 enterprise vendor environments alongside our proprietary applications, i3 Technologies captures corporate training, proof-of-concept (POC), and software implementation budgets across financial institutions, telcos, government agencies, and universities.

REVENUE STREAM	TARGET MARKET SEGMENT	COMMERCIAL MODEL & UNIT ECONOMICS	YEAR 1 PROJECTION
Stream 1: Customer & Corporate Training	Students, universities, and corporate technical teams pursuing vendor certifications (Cisco, Red Hat, AWS, Fortinet).	- Individual: Ksh 25,000/seat (3 vendor labs, 100 hrs/mo) - Professional: Ksh 75,000/seat (10 vendor labs, exam vouchers) - Enterprise: Ksh 150,000/seat (Custom labs, analytics dashboard)	Ksh 8,750,000 (120 enterprise/pro seats)
Stream 2: Labs-as-a-Service (LaaS)	System integrators, financial institutions, and software vendors needing isolated sandboxes for pre-sales POCs and dev/test.	- POC Environment: Ksh 100,000/month (30-90 day persistent lab) - Dev/Test Environment: Ksh 350,000/month - Dedicated Production Mirror: Ksh 500,000/month	Ksh 43,200,000 (Target: Ksh 69.6M total)
Stream 3: Implementation Services	Enterprise network, security, data center, and cloud migration projects unlocked via pre-validated lab environments.	- Network Infrastructure: Ksh 2,500,000 / project - Enterprise Security Architecture: Ksh 4,000,000 / project - Cloud / DC Virtualization: Ksh 5,000,000 - 7,500,000 / project	Ksh 80,000,000 (Target: Ksh 188M total)
Stream 4: Recurring Managed Services	Annual SLA agreements for Day-2 cluster operations, AI model management, security patching, and compliance monitoring.	Annual SLA retainers from Ksh 2,500,000 to Ksh 5,000,000 per enterprise client.	Ksh 25,000,000 (10 active enterprise clients)

2. UNIFIED LAB GATEWAY ARCHITECTURE & INGRESS FEDERATION

To deliver a frictionless enterprise user experience, i3 SmartLabs federates access across multiple IBM Cloud VPC accounts, on-premises Power Systems (750 VMs), and external simulator workloads via a single-pane-of-glass edge gateway:

```
# Central Ingress Topology & Federated Ingress Routing [Internet / Enterprise WAN / Campus Network] | ▼  
(Cloudflare WAF / DDoS Mitigation / Anycast DNS) Unified Edge Gateway: labs.i3technologies.co.ke (Envoy Proxy  
/ IBM Cloud ALB) | |→ Central Identity Plane: Red Hat Keycloak SSO (sso.i3technologies.co.ke/auth) | |—  
Shared Realm: i3 | Clients: evalos-spa, jupyterhub, sage, afroerp, engage, pmaas | |— Secrets: OpenBao Vault  
(MPL 2.0 Vault fork) with 2-of-3 split unseal governance | |→ Account 1 (ROKS Core Production VPC): | |—  
PostgreSQL HA (Crunchy PG0: 3 replicas + PgBouncer) | |— Workloads: i3-evalos, i3-afroerp, i3-engage, i3-  
pmaas, i3-auth, GitOps ArgoCD | |→ Account 2 (ROKS SmartLabs Compute Engine VPC): | |— Multi-Tenant  
Sandboxes: RHEL 9 UBI, Kali Linux, LocalStack, KubeVirt appliances | |— KEDA cron autoscaler: 08:00-20:00 EAT  
peak scaling; overnight resource conservation | |→ Account 3 (AI Model Gateway & GPU VPC): |— LiteLLM Proxy  
(:4000) + Langfuse Observability (langfuse.i3technologies.co.ke) |— Quantized Ollama Store (80Gi PVC):  
granite-nano, qwen-fast, coder, qwen-heavy, vision |— Burstable GPU Instances: NVIDIA A100 PCIe via GPU4YOU  
discount voucher ($1.80/hr)
```

3. THE 45-VENDOR MULTI-TENANT LAB ENVIRONMENTS MATRIX

Below is the complete inventory of all 45 vendor lab environments integrated into i3 SmartLabs, including virtualization platforms, target certifications, deployment priorities, setup costs, and operational expenses:

VENDOR	CATEGORY	VIRTUAL PLATFORM / SIMULATOR	TARGET CERTIFICATIONS	PRIORITY	EST. SETUP (KSH)	MONTHLY OPEX
1. Cisco	Networking	VIRL/CML, GNS3, IOSv	CCNA 200-301, CCNP Enterprise	HIGH	Ksh 500,000	Ksh 50,000
2. Juniper	Networking	vSRX, vMX, vQFX	JNCIA-Junos, JNCIS-ENT	HIGH	Ksh 400,000	Ksh 40,000
3. Aruba Networks	Networking	CX Simulators, AOS-CX	ACMA, ACMP, ACSP	MEDIUM	Ksh 300,000	Ksh 30,000
4. Arista Networks	Data Center	vEOS	ACE Certifications	MEDIUM	Ksh 350,000	Ksh 35,000
5. Fortinet	Security	FortiGate VM, FortiAnalyzer	NSE 4, NSE 5, NSE 7	HIGH	Ksh 450,000	Ksh 45,000
6. Check Point	Security	CloudGuard IaaS, Security GW	CCSA R81, CCSE R81	HIGH	Ksh 500,000	Ksh 50,000
7. Palo Alto	Security	VM-Series, Panorama	PCNSA, PCNSE, PCCSE	HIGH	Ksh 600,000	Ksh 60,000
8. Sophos	Security	XG Firewall Virtual	Sophos Certified Engineer	MEDIUM	Ksh 250,000	Ksh 25,000
9. F5	Load Balancing	BIG-IP Virtual Edition	F5-CA, F5-CTS	MEDIUM	Ksh 400,000	Ksh 40,000
10. Kaspersky	Endpoint Security	Security Center VM	KCSP	LOW	Ksh 150,000	Ksh 15,000
11. CrowdStrike	Endpoint Security	Falcon Platform Labs	CrowdStrike Certified	MEDIUM	Ksh 200,000	Ksh 20,000
12. Zscaler	Cloud Security	ZIA/ZPA Demo Tenants	Zscaler Certified	MEDIUM	Ksh 250,000	Ksh 25,000
13. VMware	Virtualization	Nested ESXi, VCSA, NSX-T	VCP-DCV, VCAP	HIGH	Ksh 700,000	Ksh 70,000
14. Nutanix	HCI	Nutanix CE	NCP, NCM, NCA	MEDIUM	Ksh 400,000	Ksh 40,000
15. AWS	Cloud	LocalStack Simulator	CLF, SAA, SOA, DVA, etc.	HIGH	Ksh 300,000	Ksh 30,000
16. Google Cloud	Cloud	GCP Emulator Suite	ACE, PCA, PDE	HIGH	Ksh 300,000	Ksh 30,000
17. IBM	Enterprise Systems	Power Systems LPARs	AIX, IBM i, Cloud, Watson	HIGH	Ksh 1,000,000	Ksh 100,000
18. Red Hat	Linux/OpenShift	OpenShift Cluster	RHCSA, RHCE, OpenShift Admin	HIGH	Ksh 500,000	Ksh 50,000
19. Microsoft	Cloud/Windows	Azure simulators, Windows Server	AZ-104, AZ-900, MS-900	HIGH	Ksh 400,000	Ksh 40,000
20. Oracle	Database/Cloud	Oracle Cloud, Database VMs	OCA, OCP, Cloud	MEDIUM	Ksh 500,000	Ksh 50,000
21. NetApp	Storage	ONTAP Simulators	NCDA, NCIE	MEDIUM	Ksh 350,000	Ksh 35,000

VENDOR	CATEGORY	VIRTUAL PLATFORM / SIMULATOR	TARGET CERTIFICATIONS	PRIORITY	EST. SETUP (KSH)	MONTHLY OPEX
22. Dell	Infrastructure	PowerEdge, PowerStore VMs	DCS-SA, DCS-IE	MEDIUM	Ksh 400,000	Ksh 40,000
23. HPE	Infrastructure	HPE Virtual platforms	HPE ASE, Master ASE	MEDIUM	Ksh 400,000	Ksh 40,000
24. Veeam	Backup	Veeam B&R Virtual	VMCE, VMCA	LOW	Ksh 150,000	Ksh 15,000
25. Veritas	Backup	NetBackup Virtual	Veritas Certified	LOW	Ksh 150,000	Ksh 15,000
26. Avaya	Unified Comms	Avaya Virtual platforms	ACSS, ACIS	LOW	Ksh 200,000	Ksh 20,000
27. Cisco Collaboration	Unified Comms	CUCM, Unity, Webex VMs	CCNA Collaboration	MEDIUM	Ksh 350,000	Ksh 35,000
28. SolarWinds	Monitoring	SolarWinds Virtual	SCP	LOW	Ksh 100,000	Ksh 10,000
29. Splunk	SIEM/Analytics	Splunk Enterprise Trial	Splunk Core, Power	MEDIUM	Ksh 250,000	Ksh 25,000
30. Huawei	Networking/Data Center	eNSP, FusionCompute	HCIA, HCIP, HCIE	LOW	Ksh 250,000	Ksh 25,000
31. Lenovo	Infrastructure	ThinkSystem Virtual	Data Center Solutions	LOW	Ksh 150,000	Ksh 15,000
32. Pure Storage	Storage	FlashArray//X Virtual	PFSA, PFSE	LOW	Ksh 200,000	Ksh 20,000
33. Rubrik	Backup/ Recovery	Rubrik Virtual	RCSA	LOW	Ksh 150,000	Ksh 15,000
34. Cohesity	Data Management	Cohesity Virtual	Cohesity Architect	LOW	Ksh 150,000	Ksh 15,000
35. Commvault	Backup	Commvault Virtual	CVSP	LOW	Ksh 150,000	Ksh 15,000
36. NVIDIA	AI/GPU	GPU Cloud instances	Various DLI certs	MEDIUM	Ksh 300,000	Ksh 50,000
37. Salesforce	CRM/Platform	Developer orgs	ADM 201, PD I/II	LOW	Ksh 100,000	Ksh 10,000
38. ServiceNow	ITSM	Personal Dev instances	CSA, CAD, CIS	LOW	Ksh 150,000	Ksh 15,000
39. Atlassian	Collaboration	Jira, Confluence Cloud	ACP certifications	LOW	Ksh 100,000	Ksh 10,000
40. SAP	ERP	SAP NetWeaver trial	Various SAP certs	MEDIUM	Ksh 500,000	Ksh 50,000
41. Teradata	Data Warehouse	Teradata Express	Teradata 14 Basics	LOW	Ksh 150,000	Ksh 15,000
42. MongoDB	Database	MongoDB Atlas	DBA, Developer	LOW	Ksh 100,000	Ksh 10,000
43. Mitel	Unified Comms	MiVoice Virtual	Various Mitel certs	LOW	Ksh 150,000	Ksh 15,000
44. Poly	Collaboration	Poly Virtual labs	Various Poly certs	LOW	Ksh 100,000	Ksh 10,000
45. Zoom	Video Conferencing	Zoom dev sandbox	Zoom Admin, Sales	LOW	Ksh 50,000	Ksh 5,000

4. IBM CLOUD CREDIT OPTIMIZATION & INFRASTRUCTURE BALANCING

i3 Technologies controls **three distinct IBM Cloud accounts**, each allocated **USD 3,000 per quarter starting Q4** (totaling **USD 9,000/quarter** or **USD 36,000/year**). The workload partitioning strategy below guarantees zero out-of-pocket cash infrastructure expenditure during ramp-up:

IBM CLOUD ACCOUNT	ARCHITECTURE ROLE & TARGET WORKLOADS	MONTHLY ALLOCATION	QUARTERLY BUDGET	OPTIMIZATION & GOVERNANCE CONTROLS
Account 1: Production Core & Platform SaaS	Hosts revenue-critical production SaaS: EvalOS Web, AfroERP, Engage CRM, PMaaS, Keycloak SSO, Crunchy PostgreSQL HA, and ArgoCD GitOps engine.	\$1,000 / mo	\$3,000 / Qtr	Fixed 3-node ROKS cluster (bx2-4x16); BUYVPC partner tier (60% discount). Baseline operational SLA of 99.9%.
Account 2: i3 SmartLabs Training & Compute Engine	Powers burstable student lab workloads: JupyterHub notebooks, containerized Linux/Kali sandboxes, LocalStack, and network node emulators.	\$1,000 / mo	\$3,000 / Qtr	KEDA cron autoscaling: scales up to 8 worker nodes during business/ bootcamp hours (08:00–20:00 EAT) and scales to baseline overnight.
Account 3: Model Gateway & GPU Sandbox Labs	Houses LiteLLM model proxy, Langfuse tracing, specialized watsonx AI developer sandboxes, and burstable GPU Virtual Server Instances (NVIDIA A100).	\$1,000 / mo	\$3,000 / Qtr	Leverages GPU4YOU 50% discount voucher (\$1.80/hr per A100 PCIe) to provide high-speed model fine-tuning without exhausting budget.
Total Consolidated Fabric	Tri-Account Federated SmartLabs Cloud Infrastructure	\$3,000 / mo	\$9,000 / Qtr	Net Cash Compute OpEx: Ksh 0 (\$0) during ramp-up.

5. LICENSING STRATEGY: ZERO-COST COMPUTING & PARTNER ENTITLEMENTS

To maximize gross operating margins and eliminate startup friction, i3 Technologies leverages developer subscriptions, partner Not-For-Resale (NFR) licenses, and open-source infrastructure foundations:

- **Red Hat Developer for Organizations & Partner Portal:** Grants no-cost enterprise subscriptions for up to 16 RHEL instances, complimentary OpenShift Container Platform development tiers, and Not-For-Resale (NFR) licenses for Ansible Automation Platform (AAP) and Red Hat Advanced Cluster Management (RHACM).
- **IBM Partner Plus Software Access Catalog:** Access to full software production-mirror licenses including IBM Db2 SaaS, IBM MQ, API Connect, and watsonx Orchestrate/Assistant practice licenses for demonstration and bootcamp enablement.
- **Vendor NFR Program Activation:** Formal enrollment into Fortinet Engage (FortiGate VM/FortiAnalyzer), Palo Alto NextWave (VM-Series firewalls), Check Point Partner Program (CloudGuard), and VMware Partner Connect (vSphere/ESXi NFRs) to equip 45 vendor lab pods with legitimate testing licenses.
- **Cloud Simulator Frameworks:** Utilizing LocalStack Pro (AWS), Azure Resource Manager local emulator, and GCP Google Cloud SDK test suites to deliver authentic cloud API environments with zero external egress or cloud consumption billing.
- **Open-Source Core Stack:** Kubernetes-native open-source building blocks: Crunchy Data PGO (PostgreSQL HA), Ollama + LiteLLM (LLM serving), Monaco Editor (VS Code in-browser engine), and OpenBao (Vault-compatible secret management).

Capital Expenditure Analysis for Ramp-Up: Software licensing cost during Phase 1 is **Ksh 0 (\$0)** through full utilization of Partner NFRs and Developer subscriptions. Compute costs are **100% offset** by the \$36,000/yr IBM Cloud Credits. Total cash investment required is limited to marketing, exam voucher procurement, and staff payroll.

6. INFRASTRUCTURE AUTOMATION: RED HAT ANSIBLE PLAYBOOK FRAMEWORK

Manual lab configuration does not scale to 2,500 simultaneous users. Red Hat Ansible Automation Platform (AAP) orchestrates end-to-end lifecycle provisioning, configuration drift repair, and scheduled teardown across our multi-cloud fabric:

```
--- # i3 SmartLabs Automated Student Sandbox Provisioner - name: Provision Multi-Vendor Candidate Lab
Environment hosts: localhost gather_facts: no vars: student_id: "std-kenya-0842" lab_track: "cisco-fortinet-
hybrid" session_ttl: "14400" # 4 hours tasks: - name: Create Isolated OpenShift Namespace kubernetes.core.k8s:
name: "lab-{ student_id }" kind: Namespace state: present - name: Apply Strict Network Policy & Resource Quota
kubernetes.core.k8s: definition: apiVersion: v1 kind: ResourceQuota metadata: name: student-quota namespace:
"lab-{ student_id }" spec: hard: requests.cpu: "2" requests.memory: "4Gi" limits.cpu: "4" limits.memory: "8Gi"
- name: Deploy FortiGate Virtual Appliance & Linux Terminal kubernetes.core.k8s: state: present src: "roles/
smartlabs/templates/fortigate-pod.yaml.j2" - name: Schedule Automated Teardown via KEDA/AAP Callback
ansible.builtin.shell: cmd: "echo 'oc delete ns lab-{ student_id }' | at now + 4 hours"
```

7. COMMERCIALIZATION OF PLATFORM APPLICATIONS

i3 Technologies monetizes its existing suite of proprietary applications by packaging them into standalone SaaS products and integrated modules within i3 SmartLabs:

PLATFORM APPLICATION	TECHNICAL ARCHITECTURE	COMMERCIAL PACKAGING & TARGET MARKET	PROJECTED REVENUE TRAJECTORY
EvalIOS (Exam & Skills Testing)	Next.js 14, Monaco code sandbox, Crunchy Postgres HA, LiteLLM with Qwen 7B Study Coach, and Zuri AI assistant.	Exam-as-a-Service (EaaS): B2B talent screening for tech banks and recruitment agencies. B2C practice exams for AWS, Red Hat, and IBM certifications. <i>Pricing: Ksh 1,500 - 3,500/test or Ksh 65,000/mo enterprise subscription.</i>	Year 1: Ksh 18.5M Year 2: Ksh 42.0M Year 3: Ksh 75.0M
AfroERP (Enterprise ERP & Tax)	ERPNext v15, MariaDB, Redis, integrated Safaricom Daraja STK Push / B2C, real-time KRA eTIMS invoice generation, Mfumo AI.	SME Compliance SaaS: All-in-one ERP for Kenyan retail, supply chain, and manufacturing businesses needing automated eTIMS & M-Pesa ledger sync. <i>Pricing: Ksh 6,500 - 18,500/month per business tenant.</i>	Year 1: Ksh 14.2M Year 2: Ksh 35.0M Year 3: Ksh 60.0M
Engage (Customer Communications)	Node.js/Express, Africa's Talking SMS API, Meta WhatsApp Business API, Brevo Email, Nuru AI copywriter.	Conversational CRM: Automated transactional alerts, two-way WhatsApp marketing, and payment collection reminders. <i>Pricing: Ksh 5,000/mo base platform + wholesale pass-through messaging tariffs.</i>	Year 1: Ksh 12.0M Year 2: Ksh 28.0M Year 3: Ksh 45.0M
PMaaS (Campaign Management)	NextAuth, Keycloak, Qwen 14B strategic briefing engine, voter constituency mapping, mass M-Pesa field disbursements, Dawa AI co-worker.	Political Campaign-as-a-Service: Turnkey digital operations for parliamentary, gubernatorial, and civic election campaigns across East Africa. <i>Pricing: Ksh 200,000 - 750,000/month per campaign retainer.</i>	Year 1: Ksh 25.0M Year 2: Ksh 55.0M Year 3: Ksh 80.0M
AI Lab / JupyterHub (Data Science Workbench)	JupyterHub, Open WebUI (Sage), ChromaDB vector store, on-cluster LiteLLM gateway with granite-nano, qwen-fast, and coder.	AI Workforce Incubator: Turnkey data science sandboxes for corporate upskilling, bootcamps, and hackathons. <i>Pricing: Ksh 4,000/student/mo or Ksh 350,000 per corporate training cohort.</i>	Year 1: Ksh 15.0M Year 2: Ksh 30.0M Year 3: Ksh 50.0M

8. COMPREHENSIVE 3-YEAR FINANCIAL MODEL & 90-DAY EXECUTION PLAN

Consolidating revenue from all commercial streams and platform applications yields the following financial statement (in Kenyan Shillings):

FINANCIAL STATEMENT ITEM (KSH)	YEAR 1 (FOUNDATION)	YEAR 2 (EXPANSION)	YEAR 3 (MARKET SCALE)
Customer & Corporate Training Services	6,750,000	15,000,000	25,000,000
Labs-as-a-Service (LaaS & Sandboxes)	43,200,000	80,000,000	120,000,000
Enterprise Implementation & Integration Projects	80,000,000	120,000,000	180,000,000
Managed Services & Recurring Maintenance Contracts	25,000,000	50,000,000	75,000,000
TOTAL GROSS REVENUE	154,950,000	265,000,000	400,000,000
Infrastructure, Cloud & Data Center Operations	15,000,000	20,000,000	25,000,000
Software Licenses, Simulators & Exam Vouchers	12,000,000	15,000,000	18,000,000
Engineering, Instruction & Development Personnel (10-20 staff)	25,000,000	35,000,000	45,000,000

FINANCIAL STATEMENT ITEM (KSH)	YEAR 1 (FOUNDATION)	YEAR 2 (EXPANSION)	YEAR 3 (MARKET SCALE)
Sales, Marketing & Channel Partner Enablement	8,000,000	12,000,000	15,000,000
Operations, Administration & Facilities Support	5,000,000	8,000,000	10,000,000
TOTAL OPERATING EXPENSES	65,000,000	90,000,000	113,000,000
EBITDA (Operating Profit)	89,950,000	175,000,000	287,000,000
EBITDA Margin %	58.05%	66.04%	71.75%

90-Day Tactical Execution Plan

- Month 1 (Days 1–30): Foundation & Cross-Account Transit Peering**
 - Establish VPC Peering between IBM Cloud Account 1, Account 2, and Account 3.
 - Deploy central Envoy/ALB gateway fronting Keycloak (`i3-auth`) and LiteLLM (`i3-model-gateway`).
 - Apportion ROKS workloads to begin burning the \$9,000 Q4 cloud credit allocation without cash drain.
- Month 2 (Days 31–60): Partner NFR Licensing & Automation Engine**
 - Activate vendor NFR licenses across Fortinet, Palo Alto, Cisco, and VMware partner portals.
 - Deploy Red Hat Ansible Automation Platform (AAP) playbooks for automated namespace creation and cleanup.
 - Finalize live Safaricom M-Pesa Daraja and KRA eTIMS production secrets in AfroERP and PMaaS.
- Month 3 (Days 61–90): Pilot Launch & Commercial Ramp-Up**
 - Onboard 10 enterprise pilot accounts across financial services, telecom, and academia.
 - Launch public certification tracks on EvalOS for AWS, Red Hat, and IBM watsonx.
 - Roll out outbound commercial campaigns via Engage SMS/WhatsApp for AfroERP and SmartLabs subscriptions.

Document Approved for Board Review & Implementation • i3 Technologies Limited • Executive Management Committee